Project Documentation:

Multi-Agent Orchestration for RAG Systems

W.MSCIDS_GEN03 Advanced Generative AI

GitHub Repository: `https://github.com/Trista1208/advanced_genAI.git`

Google Drive:

`https://drive.google.com/drive/folders/1Joe570BZnd4bWPcRF1sT8ERo4a3H7TBw?usp=sharing`

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Introduction

Retrieval-Augmented Generation (RAG) systems combine information retrieval with large language models to answer questions based on external knowledge sources [3]. While single-retriever RAG systems are straightforward to implement, they often struggle with diverse query types: lexical methods miss semantically similar content, while dense retrievers may fail on queries requiring exact term matching.

This project develops and evaluates a multi-agent RAG system that coordinates multiple retrieval methods to improve answer quality. The system combines three complementary retrievers: BM25 for lexical matching [5], Dense retrieval using multilingual E5 embeddings for semantic similarity [6], and GraphRAG for entity-centric and relational queries [2]. An orchestration layer decides how to combine these retrievers based on query characteristics.

The system was built and evaluated on a bilingual (English/German) corpus of ETH Zurich news articles, consisting of 817 documents in the subsample and 7,544 documents in the full corpus. All experiments were conducted on both corpus sizes to assess scalability.

The report is structured as follows. Step 1 establishes baseline retrieval performance for individual methods. Step 2 designs and compares three orchestration strategies: Waterfall with conditional fallback, Voting with fixed-weight fusion, and Confidence with query-adaptive routing. Step 3 provides comprehensive evaluation including agent complementarity analysis, failure patterns, statistical significance testing, and advanced features such as adaptive learning, explainable orchestration, adversarial robustness testing, and human-in-the-loop simulation.

1 Step 1. Baseline Setup

1.1 The Setup

The baseline system establishes a full pipeline from raw text documents to retriever construction and evaluation. It is run in a GPU-enabled Colab environment, where the project repository is cloned and both data and outputs are stored in Google Drive.

The results

The baseline setup is used to construct the following types of retrievers and measure their performance: BM25, Dense Retrieval, GraphRAG, Hybrid Retrieval, Re-Ranking (EcoRank, GTE, OpenAI, Cohere).

The performance of these retrievers are measured using the following metrics:

- **Precision@k:** the proportion of relevant documents among the top k retrieved. Higher is better.
- **Recall@k:** the proportion of all relevant documents found within the top 100 retrieved. Higher is better.
- **MRR (Mean Reciprocal Rank):** measures the average of the reciprocal ranks of the first relevant document. Higher is better.

The retriever's performance is measured on 25 bilingual benchmark questions with LLM-graded qrels (relevance greater than 0.5). The performance is tested for fixed size chunks (Figure 1).

1.2 Results on Subsample (817 documents)

BM25 Retrieval (Multilingual Baseline)

Fast lexical baselines are enhanced with BM25 pre-retrieval strategies (routing, bilingual rewriting, pseudo-relevance feedback, acronym expansion, temporal filtering). The pre-retrieval strategies are evaluated. Pseudo-relevance feedback has proven to be the best pre-retrieval strategy (Figure 1). Even though its performance on MRR is rather mediocre, it performs best for Precision@5, Precision@10 and Recall@100 for both fixed size chunks and semantic chunks. And since performance is best with fixed size chunks, BM25 with pseudo-relevance feedback with fixed chunks is chosen as the base model.

Dense Retrieval with Multilingual E5: Dense retrieval using intfloat/multilingual-E5-large-instruct with Chroma, providing semantic search through multilingual embeddings. Dense retrieval performs better with fixed size chunks than semantic chunks for all metrics except for Recall@100. It also performs better than the BM25 variants especially MRR increases significantly (Figure 1).

GraphRAG: An entity/event graph is built, clustered into Leiden communities, and condensed LLM summaries are generated for retrieval. Although GraphRAG-C2 achieves the best performance compared to the other two GraphRAG models, it performs worse than dense retrieval in all metrics (Figure 1).

Hybrid Retrieval: For hybrid retrieval late fusion of BM25, dense retrieval and GraphRAG is used. Three different fusion operators (weighted sum, z-score sum and Reciprocal Rank Fusion) are implemented and evaluated. With a recip rank of 0.654, z-score sum achieves best performance across fusion operators. It is configured best with dense retrieval (0.60) and GraphRAG (0.40). The hybrid retrieval approach achieves best performance in all metrics, except for Precision@5 (Figure 1).

Re-Ranking: The fine-tuned hybrid retrieval model, which achieves the best performance, is used as baseline for the re-ranking. For the re-ranking EcoRank, GTE, OpenAI and Cohere are tested and evaluated (Figure 2). All of the re-ranking methods increased performance compared to traditional retrieval methods. Cohere performs best, achieving highest scores in all metrics, except for MRR.

1.3 Results on Full Corpus (7,544 documents)

To validate scalability, the baseline retrievers (BM25, Dense, GraphRAG) are also evaluated on the full corpus, which is approximately 9 times larger than the subsample.

As expected, all metrics decrease substantially when scaling from the subsample to the full corpus (Figure 3). This is a natural consequence of the increased search space, making the retrieval task more challenging. Dense retrieval maintains its lead in MRR (0.166) and Recall@100 (0.128), confirming that the relative performance trends observed on the subsample generalize to the full dataset.

1.4 Conclusion

For maximizing retrieval quality, reranking provides substantial benefits over traditional retrieval methods. The hybrid-tuned retriever itself is a strong baseline, demonstrating the power of late fusion. When considering deployment, the trade-off between reranking quality and latency must be carefully evaluated. Although Cohere scores best, it comes with high latency. The consistency of relative performance rankings between subsample and full corpus validates that the subsample is representative for method selection and hyperparameter tuning.

2 Step 2. Multi-Agent System Design

2.1 Overview

Step 2 implements a multi-agent orchestration layer that coordinates multiple retrieval systems to improve retrieval quality and answer synthesis. We designed and evaluated three orchestration strategies: Waterfall, Voting, and Confidence-based routing. Each strategy represents a different approach to combining retriever outputs, balancing quality, efficiency, and adaptability to query types.

2.2 System Architecture

The multi-agent system combines three retriever agents. The BM25 Agent performs bilingual lexical retrieval with separate English and German indices, using Facebook's M2M100 model for automatic query translation. The Dense Agent provides semantic retrieval using multilingual E5 embeddings with the Chroma vector store. The GraphRAG Agent performs graph-based retrieval using Leiden community clustering and LLM-generated summaries.

Two supporting agents complete the architecture. The Query Classifier Agent uses a lightweight ‘google/flan-t5-base‘ model to categorize queries as FACTOID, SEMANTIC, or HYBRID, enabling dynamic weight adjustment. The Answer Synthesizer Agent generates final answers using ‘Mistral-7B-Instruct-v0.2‘ based on retrieved context.

All orchestration strategies combine retriever outputs using weighted Reciprocal Rank Fusion (RRF) [1]. This fusion method assigns each document a score based on its rank position in each retriever's output, weighted by retriever importance. Documents ranked higher receive proportionally more credit, and the weights allow prioritising certain retrievers over others. A smoothing constant prevents excessive emphasis on top-ranked documents.

2.3 Orchestration Strategies

Waterfall Strategy implements sequential routing with conditional fallback. It initially retrieves using only BM25 and Dense with weights favoring BM25 (1.2) over Dense (1.0). A critic agent then evaluates the top result's query-term overlap. If this overlap falls below 5%, indicating poor lexical match, GraphRAG is activated as fallback with adjusted weights. This strategy optimizes for efficiency by avoiding expensive GraphRAG calls when lexical and semantic retrieval prove sufficient.

Voting Strategy performs parallel retrieval with fixed-weight fusion. All three retrievers run simultaneously with predetermined weights: BM25 at 1.2, Dense at 1.0, and GraphRAG at 0.6. Results are fused using weighted RRF without any conditional logic. This strategy maximizes coverage by always utilising all available retrieval signals.

Confidence Strategy dynamically adjusts weights based on query classification. The Query Classifier Agent first categorizes the query, then weights are set accordingly. For FACTOID queries involving names, numbers, and dates, BM25 receives higher weight for precise term matching. For SEMANTIC queries involving explanations and concepts, Dense retrieval is boosted for better conceptual understanding. For HYBRID queries mixing both types, balanced weights are applied across all retrievers.

2.4 Architectural comparison: subsample vs full corpus

2.4.1 Retriever configuration differences

The most significant change concerns the BM25 retriever. On the subsample, BM25 uses Query Expansion via Pseudo-Relevance Feedback (PRF) [4], which improves recall by expanding queries with frequent terms from top initial results. On the full corpus, this expansion is removed in favor of baseline BM25. This decision reflects the trade-off between recall improvement and computational cost, PRF requires two retrieval passes and risks topic drift from noisy expansion terms when the corpus is larger.

The pre-retrieval depth also differs between configurations. The subsample uses dynamic pre-k calculated as the maximum of 30 or ten times the requested top-k, adapting to query needs. The full corpus uses a fixed pre-k of 120, which ensures consistent deep retrieval regardless of query characteristics. This change simplifies system behavior and guarantees sufficient candidates for effective fusion in the larger search space.

Additionally, BM25 scoring differs due to implementation constraints. The subsample uses native BM25Okapi scores, while the full corpus uses rank-based scoring where each document receives a score equal to the total count minus its rank position. This change accommodates the LangChain BM25Retriever interface, which does not expose raw scores, while maintaining consistent scale for fusion with other retrievers.

2.4.2 Orchestration Weight Adjustments

The Confidence strategy's weights were adjusted for the full corpus, particularly for FACTOID queries. We increased GraphRAG's weight substantially from 0.5 to 0.8, while slightly decreased BM25 from 1.4 to 1.3 and increased Dense from 0.9 to 1.0. This change underscores the growing role of GraphRAG in larger search spaces, where relevant factual information is spread across many documents, making entity-based graph traversal increasingly important for identifying and connecting the required evidence. The weights for SEMANTIC and HYBRID query types remain unchanged between configurations, as the relationship between conceptual queries and retriever capabilities does not fundamentally change with corpus size.

2.5 Evaluation Results

2.5.1 Answer Coverage (Hit Rate)

Hit rate measures semantic answer coverage (how often the retrieved context contains tokens from the gold answer, requiring at least two-token overlap).

On the subsample (Figure 4), Confidence achieves the highest hit rate at 0.920, followed by Voting at 0.900 and Waterfall at 0.880. This demonstrates the value of query-adaptive routing when the search space is manageable.

On the full corpus (Figure 5), Voting takes the lead with 0.940, while Confidence maintains 0.920 and Waterfall remains at 0.880. We suggest that consistent multi-retriever fusion provides more robust coverage when the search space is larger, as adaptive routing may miss relevant documents that don't fit neatly into query type categories.

2.5.2 Formal IR Metrics

On the subsample (Figure 6), Waterfall achieves the best MRR (0.447), indicating it effectively places the most relevant document at the top. However, Voting leads in precision and ranking quality metrics (P@10: 0.342, nDCG@10: 0.350).

On the full corpus (Figure 7), Voting shows higher values across all metrics with the best MRR (0.189), precision (P@5: 0.083, P@10: 0.096), and ranking quality (nDCG@10: 0.104). Importantly, Voting shows the smallest performance degradation when scaling, that demonstrates robustness to corpus size increase.

2.6 Qualitative Analysis: Answer Synthesis Examples

The following examples illustrate how different orchestration strategies affect answer quality across both datasets.

Example 1: Factual Query "Who were the rectors of ETH between 2017 and 2022?"

On the subsample, all strategies either fail to find the answer or produce incorrect responses (Figure 8). On the full corpus, both Voting and Confidence correctly identify Sarah Springman as rector from 2017 until her departure in January 2022, while Waterfall still returns "NOT FOUND IN CONTEXT" (Figure 9). This demonstrates how the larger document collection contains more relevant content, and multi-retriever fusion helps surface it effectively.

Example 2: Domain-Specific Query "Does ETH organize any competitions?"

The gold answer mentions various competitions including the notable Cybathlon. On the subsample, responses reference peripheral events like a photo contest or Olympic athletes (Figure 10). On the full corpus, Confidence successfully retrieves information about the Cybathlon, which directly matches the gold answer's most prominent example (Figure 11). This illustrates how query-adaptive routing can surface highly relevant but semantically distinct content when the retrieval space is larger.

2.7 Conclusions

Voting achieved the best results across all experiments and scaled better from subsample to full corpus. Waterfall performed well on the subsample but showed weaker results on the full corpus. The conditional GraphRAG activation saves computation time, but sometimes skips relevant documents. Confidence did not perform better than simpler approaches despite its adaptive design. The query classifier was not accurate enough to provide consistent improvements. The architectural adaptations for the full corpus were successful. Removing Query Expansion simplified the BM25 pipeline without hurting performance. Increasing GraphRAG weights helped retrieve relevant documents from the larger collection.

3 Step 3. Core Evaluation

3.1 Evaluation Methodology

A broad set of Information Retrieval (IR) metrics was used to evaluate system performance. Precision (@1, @3, @5, @10) measures relevance among top-ranked documents, Recall (@5, @10, @100) captures coverage of relevant documents, MRR reflects the rank of the first relevant result, and NDCG (@5, @10) assesses ranking quality. System efficiency was evaluated using average, P95, and P99 latency, as well as end-to-end processing time. The evaluation pipeline executes retrieval for each query, records per-query latency, computes IR metrics using pytrec_eval, and aggregates results via macro-averaging.

3.2 Quantitative Evaluation Results

3.2.1 Performance on Subsample Corpus

On the subsample corpus of 817 documents, the Waterfall strategy achieved the highest MRR of 0.447, followed by Voting at 0.421 and Confidence at 0.403 (Figure 12). This indicates that on smaller corpora, the conditional GraphRAG fallback mechanism in Waterfall provides effective coverage when initial retrieval is insufficient.

Regarding efficiency, the Confidence strategy exhibited the lowest average latency at 3.08 seconds, approximately 2.5 times faster than Waterfall (7.78 seconds) and 1.8 times faster than Voting (5.66 seconds) (Figure 12). The efficiency advantage of Confidence stems from its intelligent query routing that assigns appropriate weights based on query classification, avoiding unnecessary retrieval from all agents.

Precision metrics showed comparable performance across strategies at higher cutoffs (Figure 12). Voting achieved the highest P@10 (0.342) and NDCG@10 (0.350), suggesting better ranking quality when more results are considered. Recall@100 was similar across all strategies (approximately 0.28-0.30), indicating that all approaches can retrieve most relevant documents given sufficient depth.

3.2.2 Performance on Full Corpus

Scaling to the full corpus of 7,544 documents revealed significant performance changes (Figure 13). The Confidence strategy emerged as the best performer with MRR of 0.205, followed by Voting at 0.190 and Waterfall at 0.161. This reversal from subsample results demonstrates that query-adaptive weight assignment becomes more important as corpus size increases.

The efficiency characteristics also changed substantially. Waterfall became the fastest strategy at 3.05 seconds average latency, followed closely by Confidence at 3.36 seconds (Figure 13). This improvement occurred because Waterfall's overlap criterion is more easily satisfied on the larger corpus, reducing the frequency of GraphRAG fallback invocations. Voting exhibited considerably higher latency due to the increased cost of GraphRAG retrieval over a larger knowledge graph.

All strategies showed lower precision and recall on the full corpus, which is expected given the larger search space. However, the relative performance differences became more noticeable. Confidence and Voting achieved P@5 of approximately 0.10-0.12, while Waterfall dropped to 0.058, indicating that Waterfall's conditional approach may miss relevant documents when the overlap threshold is not triggered (Figure 13).

3.3 Qualitative Analysis

3.3.1 Agent Complementarity

The AgentComplementarityAnalyzer examined overlap among BM25, Dense retrieval, and GraphRAG agents across 15 test queries (Figure 14). On the subsample corpus, only 1.22% of retrieved documents were shared by all three agents, indicating strong complementarity. BM25-only documents accounted for 37.95%, reflecting BM25's strength in exact keyword matching. Graph-only documents comprised 33.05%, demonstrating GraphRAG's ability to retrieve relational and entity-centric information. Dense-only documents represented 14.23%, highlighting semantic retrieval's contribution for queries using different terminology than source documents.

On the full corpus, the distribution became more balanced: BM25-only at 30.34%, Dense-only at 29.99%, and Graph-only at 30.60%, with only 0.30% overlap among all three. This equalisation suggests that Dense retrieval becomes proportionally more valuable on larger corpora where semantic matching helps bridge vocabulary gaps across a wider range of documents.

These complementarity results validate the multi-agent fusion approach: each retriever contributes substantially unique documents, and combining their results through RRF captures information that any single agent would miss.

3.3.2 Failure Pattern Analysis

The FailureAnalyzer identified queries with NDCG@10 below 0.5. On the subsample, failure counts ranged from 16 to 18 across strategies, while on the full corpus this increased to 20-22 failures. The average query length among failures was approximately 8.3-8.5 words, indicating no strong correlation between query length and retrieval difficulty.

Question type distribution among failures was dominated by "What" questions (7 failures), followed by "Who" and "How" questions (4-5 each). Recurring failure patterns included queries requiring enumeration of multiple entities (such as listing all ETH rectors between specific years), queries about specific quotes or statements attributed to named individuals, and queries about niche projects or initiatives with limited textual coverage in the corpus.

Common failure examples persisted across both corpus sizes and all strategies. Queries like "Who are famous ETH alumni?" require comprehensive enumeration across many documents, while queries about specific projects like "What is e-Sling?" demand retrieval of documents that may use varying terminology or abbreviations.

3.4 System Efficiency and Statistical Analysis

Latency measurements reveal distinct efficiency profiles across strategies and corpus sizes (Figure 15). On the subsample, the Confidence strategy achieved a 2.5x speedup over Waterfall due to intelligent query routing that avoids unnecessary agent invocations. On the full corpus, this relationship partially inverted: Waterfall became fastest because its overlap criterion prevented most GraphRAG calls, while Voting's parallel retrieval from all agents incurred the highest cost.

Statistical significance testing using paired t-tests at $\alpha = 0.05$ indicated that MRR differences between strategies were not statistically significant on either corpus (Figure 16). On the subsample, p-values ranged from 0.36 to 0.63 for all pairwise comparisons. On the full corpus, the Voting versus Confidence comparison approached significance ($p = 0.065$), suggesting that with a larger query set, the Confidence strategy's advantage might reach statistical significance.

The lack of significance is attributed to the small sample size of 25 queries, high per-query variance in retrieval outcomes, and modest absolute effect sizes. Despite this, the consistent direction of results across multiple metrics supports practical conclusions about strategy selection.

4 Step 4: Advanced and Bonus Features

4.1 Adaptive Orchestration with Reinforcement Learning

An ‘AdaptiveOrchestrator‘ based on Q-learning [7] was implemented to overcome limitations of static weight assignment. Query types (factoid, semantic, balanced) form the state space, while orchestration strategies (bm25_heavy, dense_heavy, graph_heavy, balanced) constitute the action space. Rewards are derived from retrieval success measured by answer overlap, and learning proceeds via an epsilon-greedy policy with learning rate 0.1 and discount factor 0.9.

Training on 20 queries revealed different optimal strategies depending on corpus size (Figure 17). On the subsample, the system learned to prefer graph_heavy strategies for factoid queries and balanced approaches for semantic queries, achieving 75% success rate and 0.775 average feedback score. On the full corpus, the system converged toward balanced strategies across all query types, with reduced success rate (30%) and lower feedback (0.370).

The learned Q-values on the subsample showed highest values for dense_heavy on factoid queries (0.587) and graph_heavy on balanced queries (0.731). On the full corpus, balanced strategies dominated with values around 0.30-0.48 across query types. This shift indicates that as corpus size increases, no single agent provides consistently superior results, making balanced fusion more robust.

4.2 Explainable Orchestration

The ‘ExplainableOrchestrator‘ provides transparent rationales for orchestration decisions by analysing query features, classifying query type, assigning agent weights, and documenting the retrieval and fusion process. For each query, the system outputs feature analysis (length, presence of digits, named entities, question words), determined query complexity, weight assignments with explanations, and a step-by-step decision flow.

For example, the query ”Who was president of ETH in 2003?” is classified as factoid based on the presence of a specific date, resulting in BM25 weight of 1.4, Dense weight of 0.9, and Graph weight of 0.5. The rationale explicitly states that exact matching is prioritized for queries containing specific identifiers.

4.3 Adversarial Query Evaluation

An ‘AdversarialQueryGenerator‘ was used to test system robustness under challenging conditions, including ambiguous queries, code-switched English–German queries, and paraphrased queries.

On the subsample, paraphrased queries achieved 75% success rate across strategies, code-switched queries reached 40%, and ambiguous queries showed 0-50% success with only Voting handling them partially (Figure 18). On the full corpus, performance degraded: paraphrased queries dropped to 50% success, while code-switched (40%) and ambiguous (0%) remained similar (Figure 19).

The degradation for paraphrased queries on the full corpus suggests that semantic matching becomes more difficult when relevant documents are distributed among a larger pool of candidates. All strategies showed identical adversarial performance on the full corpus, indicating that robustness challenges are shared across orchestration approaches rather than being strategy-specific.

4.4 Human-in-the-Loop Simulation

A ‘HumanInTheLoopSimulator‘ was implemented to model iterative refinement using simulated human feedback. Retrieved results are scored using lenient matching criteria based on exact matches, entity overlap, and keyword coverage. Feedback is used to reinforce successful strategies and penalize failures, enabling adaptive learning over multiple iterations.

On the subsample, 16 interactions produced 9 approvals and 7 rejections, with average feedback score of 0.584 and final success rate of 100% after iterative refinement (Figure 20). On the full corpus, 12 interactions yielded 10 approvals and only 2 rejections, with higher average feedback (0.694) and 100% success rate. The reduced rejection rate on the full corpus may reflect the broader document coverage providing more opportunities for partial matches.

Both simulations demonstrated that iterative human feedback can recover from initial retrieval failures, achieving perfect task completion within 3 iterations maximum. This validates the human-in-the-loop approach for production deployment where user feedback can continuously improve system performance.

5 Discussion and Conclusions

The evaluation demonstrates that orchestration strategy effectiveness depends significantly on corpus scale. On smaller corpora, the Waterfall strategy's conditional GraphRAG fallback provides the best MRR, while on larger corpora, the Confidence strategy's query-adaptive weight assignment scales more smoothly. The Voting strategy provides consistent middle-ground performance but at higher computational cost.

Strong complementarity among BM25, Dense, and GraphRAG agents validates the multi-agent fusion approach. Each retriever contributes approximately 30-38% unique documents, with minimal three-way overlap (0.3-1.2%). This complementarity becomes more balanced on larger corpora, where Dense retrieval's semantic matching becomes proportionally more valuable.

The advanced features demonstrate pathways for system improvement. Adaptive Q-learning captures non-intuitive strategy preferences but requires corpus-specific training. Explainable orchestration supports debugging and trust-building for production deployment. Adversarial evaluation reveals robustness gaps, particularly for ambiguous queries that lack specific entities. Human-in-the-loop simulation shows that iterative feedback can recover from initial retrieval failures and improve task completion.

For practical deployment, the Confidence strategy offers the best balance of effectiveness and efficiency on larger corpora, but its performance may vary due to instability in the query classifier. Its query classification overhead is minimal compared to the savings from intelligent agent routing, and its MRR degradation is the lowest among evaluated strategies. Future work should focus on improving query classification accuracy, expanding adversarial robustness, and integrating real human feedback for continuous learning.

6 Appendix

	MRR	Precision@5	Precision@10	Recall@100	nDCG@10
BM25	0.379	0.258	0.212	0.210	0.230
Routing	0.404	0.267	0.229	0.231	0.248
Rewrite	0.306	0.200	0.162	0.210	0.171
QE	0.401	0.292	0.288	0.272	0.284
Spell	0.431	0.242	0.233	0.209	0.250
Temp	0.408	0.282	0.232	0.211	0.251
Dense	0.541	0.350	0.304	0.265	0.354
Hybrid-tuned	0.654	0.342	0.321	0.294	0.388
GraphRAG-C0	0.283	0.133	0.137	0.211	0.142
GraphRAG-C1	0.485	0.267	0.246	0.215	0.276
GraphRAG-C2	0.510	0.300	0.246	0.187	0.294

Figure 1: Subsample retriever results on fixed size chunks

	MRR	P@5	P@10	R@100	nDCG@10	latency[s]
Hybrid-tuned	0.654	0.342	0.321	0.294	0.388	0.00
EcoRank	0.662	0.408	0.379	0.294	0.433	0.19
GTE	0.678	0.400	0.321	0.294	0.397	0.28
OpenAI	0.663	0.425	0.404	0.294	0.453	76.76
Cohere	0.666	0.483	0.417	0.294	0.472	4.95

Figure 2: Subsample reranking results

	MRR	Precision@5	Precision@10	Recall@100	nDCG@10
BM25	0.153	0.083	0.096	0.080	0.087
Routing	0.185	0.092	0.104	0.084	0.097
Rewrite	0.092	0.042	0.050	0.049	0.046
QE	0.103	0.058	0.079	0.063	0.068
Spell	0.156	0.067	0.083	0.082	0.078
Temp	0.165	0.091	0.105	0.087	0.095
Dense	0.166	0.058	0.067	0.128	0.071
GraphRAG-C0	0.029	0.008	0.008	0.012	0.007
GraphRAG-C1	0.051	0.017	0.025	0.014	0.021
GraphRAG-C2	0.116	0.050	0.033	0.045	0.037

Figure 3: Full corpus retriever results on fixed size chunks

```
Answer-in-context (ALL questions): 100%|██████████| 50/50 [02:33<00:00,  3.07s/it]RESULTS (ALL questions):
Waterfall hit rate:  0.880
Voting hit rate:     0.900
Confidence hit rate: 0.920
```

Figure 4: Subsample Hit Rate

```
Answer-in-context (ALL questions): 100%|██████████| 50/50 [02:40<00:00,  3.22s/it]RESULTS (ALL questions):
Waterfall hit rate:  0.880
Voting hit rate:     0.940
Confidence hit rate: 0.920
```

Figure 5: Full corpus Hit Rate

```
Waterfall: 100%|██████████| 25/25 [00:06<00:00,  3.59it/s]
Voting: 100%|██████████| 25/25 [00:07<00:00,  3.47it/s]
Confidence: 100%|██████████| 25/25 [01:00<00:00,  2.42s/it]
```

	recip_rank	P_5	P_10	recall_100	ndcg_cut_10
Waterfall	0.447	0.308	0.296	0.283	0.311
Voting	0.421	0.325	0.342	0.298	0.350
Confidence	0.403	0.325	0.321	0.298	0.329

Figure 6: Orchestration performance on subsample (817 documents)

```
Waterfall: 100%|██████████| 25/25 [00:06<00:00,  3.77it/s]
Voting: 100%|██████████| 25/25 [00:06<00:00,  3.69it/s]
Confidence: 100%|██████████| 25/25 [00:38<00:00,  1.53s/it]
```

	recip_rank	P_5	P_10	recall_100	ndcg_cut_10
Waterfall	0.151	0.050	0.046	0.128	0.051
Voting	0.189	0.083	0.096	0.121	0.104
Confidence	0.139	0.067	0.071	0.123	0.064

Figure 7: Orchestration performance on full corpus (7544 documents)

```
################################################################################
QUESTION 2: who were the rectors of eth between 2017 and 2022?
GOLD ANSWER: sarah springman, günther dissertori.
Setting `pad_token_id` to `eos_token_id`:2 for open-end generation.
------------------------------------------------------------
STRATEGY: WATERFALL
GENERATED ANSWER:
NOT FOUND IN CONTEXT. The context does not provide information about the rectors of ETH Zurich between 2017 and 2022.
------------------------------------------------------------
STRATEGY: VOTING
GENERATED ANSWER:
NOT FOUND IN CONTEXT. The context does not provide information about the rectors of ETH Zurich between 2017 and 2022.
Setting `pad_token_id` to `eos_token_id`:2 for open-end generation.
Setting `pad_token_id` to `eos_token_id`:2 for open-end generation.
------------------------------------------------------------
STRATEGY: CONFIDENCE
GENERATED ANSWER:
NOT FOUND IN CONTEXT. The context does not provide information about the rectors of ETH Zurich between 2017 and 2022.
```

Figure 8: Example 1 subsample

```
################################################################################
QUESTION 2: who were the rectors of eth between 2017 and 2022?
GOLD ANSWER: sarah springman, günther dissertori.
Setting `pad_token_id` to `eos_token_id`:2 for open-end generation.
------------------------------------------------------------
STRATEGY: WATERFALL
GENERATED ANSWER:
NOT FOUND IN CONTEXT. The context does not provide information about who the rectors of ETH Zurich were between 2017 and 2022.
------------------------------------------------------------
STRATEGY: VOTING
GENERATED ANSWER:
Sarah Springman was the rector of ETH Zurich from 2017 to 2022.
Setting `pad_token_id` to `eos_token_id`:2 for open-end generation.
Setting `pad_token_id` to `eos_token_id`:2 for open-end generation.
------------------------------------------------------------
STRATEGY: CONFIDENCE
GENERATED ANSWER:
Sarah Springman was the rector of ETH Zurich between 2017 and her departure at the end of January 2022.
```

Figure 9: Example 1 full corpus

```
################################################################################
QUESTION 48: does eth organize any competitions?
GOLD ANSWER: eth zurich organizes a number of competitions for the public good, addressing youth, aspiring researchers, sports, innovators, and entrepreneurs. what area would you like to explore? d
Setting `pad_token_id` to `eos_token_id`:2 for open-end generation.
--------------------------------------------------------------
STRATEGY: WATERFALL
GENERATED ANSWER:
NOT FOUND IN CONTEXT. The context does not mention any information about competitions organized by ETH Zurich.
--------------------------------------------------------------
STRATEGY: VOTING
GENERATED ANSWER:
Yes, according to the context, Eth Zurich students are competing in the Olympic games. Specifically named are athletes Julie Derron (triathlon), Annina Fahr (athletics), Geraldine Frey (athletics),
Setting `pad_token_id` to `eos_token_id`:2 for open-end generation.
Setting `pad_token_id` to `eos_token_id`:2 for open-end generation.
--------------------------------------------------------------
STRATEGY: CONFIDENCE
GENERATED ANSWER:
Yes, according to the context, Eth Zurich students are competing in the Olympics.
```

Figure 10: Example 2 subsample

```
################################################################################
QUESTION 48: does eth organize any competitions?
GOLD ANSWER: eth zurich organizes a number of competitions for the public good, addressing youth, aspiring researchers, sports, innovators, and entrepreneurs. what area would you like to explore? d
Setting `pad_token_id` to `eos_token_id`:2 for open-end generation.
--------------------------------------------------------------
STRATEGY: WATERFALL
GENERATED ANSWER:
Yes, Eth Zurich organizes the Eth Zurich Latsis Prize, an annual competition to honor outstanding young scientists from all fields at Eth Zurich.
--------------------------------------------------------------
STRATEGY: VOTING
GENERATED ANSWER:
Yes, according to the context, Eth does organize competitions such as the game mentioned in the text where participants can win prizes from the FIFA museum, the Eth store, and a Keep Cup. Eth studer
Setting `pad_token_id` to `eos_token_id`:2 for open-end generation.
Setting `pad_token_id` to `eos_token_id`:2 for open-end generation.
--------------------------------------------------------------
STRATEGY: CONFIDENCE
GENERATED ANSWER:
Yes, Eth Zurich organizes competitions such as the Cybathlon for people with disabilities who use assistive technology, and the game described in the context which is dedicated to robotics, gaming
```

Figure 11: Example 2 full corpus

	Strategy	recip_rank	P_1	P_3	P_5	P_10	recall_5	recall_10	recall_100	ndcg_cut_5	ndcg_cut_10	Avg_Latency(s)	P95_Latency(s)
0	Waterfall	0.4469	0.2917	0.2778	0.3083	0.2958	0.0654	0.0826	0.2828	0.3110	0.3106	7.7822	13.0445
1	Voting	0.4211	0.2083	0.3611	0.3250	0.3417	0.0649	0.0995	0.2979	0.3259	0.3501	5.6574	5.8829
2	Confidence	0.4029	0.1667	0.3194	0.3250	0.3208	0.0685	0.0952	0.2977	0.3165	0.3288	3.0849	3.1130

Figure 12: Orchestration performance on subsample (817 documents)

	Strategy	recip_rank	P_1	P_3	P_5	P_10	recall_5	recall_10	recall_100	ndcg_cut_5	ndcg_cut_10	Avg_Latency(s)	P95_Latency(s)
0	Waterfall	0.1614	0.0833	0.0694	0.0583	0.0458	0.0030	0.0104	0.1277	0.0622	0.0525	3.0473	5.7500
1	Voting	0.1901	0.0417	0.1111	0.1000	0.1000	0.0042	0.0511	0.1293	0.0935	0.1072	41.4482	28.1446
2	Confidence	0.2045	0.0417	0.1111	0.1167	0.1000	0.0046	0.0511	0.1272	0.1081	0.1113	3.3635	3.2846

Figure 13: Orchestration performance on full corpus (7544 documents)

```
================================================================================
AGENT COMPLEMENTARITY SUMMARY
================================================================================
```

	query	total_unique	all_three_pct	bm25_only_pct	dense_only_pct	graph_only_pct
0	who was president of eth in 2003?	24	0.000000	41.666667	16.666667	37.500000
1	who were the rectors of eth between 2017 and 2...	23	0.000000	39.130435	13.043478	39.130435
2	who at eth received erc grants?	15	13.333333	20.000000	6.666667	20.000000
3	when did the insight get to mars?	21	0.000000	42.857143	9.523810	28.571429
4	what did prof. schubert say about flying?	24	0.000000	41.666667	16.666667	37.500000
5	what is e-sling?	25	0.000000	40.000000	20.000000	40.000000
6	who are famous eth alumni?	23	0.000000	43.478261	13.043478	34.782609
7	who at eth currently works on research regardi...	21	0.000000	33.333333	19.047619	28.571429
8	how do alpine plants respond to climate change?	21	0.000000	42.857143	4.761905	33.333333
9	how would you make ferzilizer without carbon em...	20	5.000000	35.000000	15.000000	30.000000
10	what research is eth famous for?	22	0.000000	36.363636	18.181818	31.818182
11	how much of eth's electricity consumpzon is du...	23	0.000000	43.478261	13.043478	34.782609
12	what are pseudocereals and who does research o...	25	0.000000	40.000000	20.000000	40.000000
13	who is working on methods for targeted cancer ...	18	0.000000	27.777778	11.111111	22.222222
14	how is eth research inveszgazng methods to avo...	24	0.000000	41.666667	16.666667	37.500000

```
Average Overlap Statistics:
  All 3 agents: 1.22%
  BM25 only: 37.95%
  Dense only: 14.23%
  Graph only: 33.05%
```

```
================================================================================
AGENT COMPLEMENTARITY SUMMARY
================================================================================
```

	query	total_unique	all_three_pct	bm25_only_pct	dense_only_pct	graph_only_pct
0	who was president of eth in 2003?	24	0.000000	33.333333	25.000000	25.000000
1	who were the rectors of eth between 2017 and 2...	25	0.000000	32.000000	28.000000	28.000000
2	who at eth received erc grants?	22	4.545455	22.727273	22.727273	31.818182
3	when did the insight get to mars?	23	0.000000	34.782609	21.739130	21.739130
4	what did prof. schubert say about flying?	27	0.000000	29.629630	33.333333	33.333333
5	what is e-sling?	26	0.000000	30.769231	30.769231	30.769231
6	who are famous eth alumni?	25	0.000000	32.000000	28.000000	28.000000
7	who at eth currently works on research regardi...	28	0.000000	28.571429	35.714286	35.714286
8	how do alpine plants respond to climate change?	24	0.000000	33.333333	25.000000	25.000000
9	how would you make ferzilizer without carbon em...	27	0.000000	29.629630	33.333333	33.333333
10	what research is eth famous for?	27	0.000000	29.629630	33.333333	33.333333
11	how much of eth's electricity consumpzon is du...	28	0.000000	28.571429	35.714286	35.714286
12	what are pseudocereals and who does research o...	28	0.000000	28.571429	35.714286	35.714286
13	who is working on methods for targeted cancer ...	26	0.000000	30.769231	30.769231	30.769231
14	how is eth research inveszgazng methods to avo...	26	0.000000	30.769231	30.769231	30.769231

```
Average Overlap Statistics:
  All 3 agents: 0.30%
  BM25 only: 30.34%
  Dense only: 29.99%
  Graph only: 30.60%
```

Figure 14: Agent complementarity analysis results on the subsample (left) and the full corpus (right)

```
Waterfall:
  Avg Latency: 7.7822s
  P95 Latency: 13.0445s
  Total Time: 194.55s

Voting:
  Avg Latency: 5.6574s
  P95 Latency: 5.8829s
  Total Time: 141.43s

Confidence:
  Avg Latency: 3.0849s
  P95 Latency: 3.1130s
  Total Time: 77.12s

3. KEY FINDINGS:
  - Best MRR: Waterfall (0.4469)
  - Fastest: Confidence (3.0849s)
```

```
Waterfall:
  Avg Latency: 3.0473s
  P95 Latency: 5.7500s
  Total Time: 76.18s

Voting:
  Avg Latency: 41.4482s
  P95 Latency: 28.1446s
  Total Time: 1036.20s

Confidence:
  Avg Latency: 3.3635s
  P95 Latency: 3.2846s
  Total Time: 84.09s

3. KEY FINDINGS:
  - Best MRR: Confidence (0.2045)
  - Fastest: Waterfall (3.0473s)
```

Figure 15: Efficiency metrics on the subsample (left) and on the full corpus (right)

```
=====================================
STATISTICAL SIGNIFICANCE TESTS
=====================================

Waterfall vs Voting:
  Mean Difference: 0.0258
  t-statistic: 0.4930
  p-value: 0.6267
  Significant (α=0.05): NO

Waterfall vs Confidence:
  Mean Difference: 0.0440
  t-statistic: 0.9309
  p-value: 0.3616
  Significant (α=0.05): NO

Voting vs Confidence:
  Mean Difference: 0.0182
  t-statistic: 0.7401
  p-value: 0.4667
  Significant (α=0.05): NO
```

```
=====================================
STATISTICAL SIGNIFICANCE TESTS
=====================================

Waterfall vs Voting:
  Mean Difference: -0.0287
  t-statistic: -0.4901
  p-value: 0.6287
  Significant (α=0.05): NO

Waterfall vs Confidence:
  Mean Difference: -0.0431
  t-statistic: -0.7225
  p-value: 0.4773
  Significant (α=0.05): NO

Voting vs Confidence:
  Mean Difference: -0.0144
  t-statistic: -1.9383
  p-value: 0.0650
  Significant (α=0.05): NO
```

Figure 16: Statistical significance testing on the subsample (left) and on the full corpus (right)

```
FACTOID:
  dense_heavy: 0.5867
  graph_heavy: 0.1800

BALANCED:
  graph_heavy: 0.7313
  balanced: 0.2000

SEMANTIC:
  balanced: 0.4880
  dense_heavy: 0.0200
=========================================

=========================================
TRAINING SUMMARY
=========================================

Strategy Usage:
strategy
graph_heavy    9
dense_heavy    7
balanced       4
Name: count, dtype: int64

Average Feedback: 0.775
Success Rate (answer found): 75.0%
```

```
FACTOID:
  balanced: 0.2952
  bm25_heavy: 0.0360
  dense_heavy: 0.0200
  graph_heavy: 0.0200

BALANCED:
  balanced: 0.4768
  bm25_heavy: 0.0200

SEMANTIC:
  bm25_heavy: 0.1928
  balanced: 0.0200
=========================================

=========================================
TRAINING SUMMARY
=========================================

Strategy Usage:
strategy
balanced       12
bm25_heavy      6
dense_heavy     1
graph_heavy     1
Name: count, dtype: int64

Average Feedback: 0.370
Success Rate (answer found): 30.0%
```

Figure 17: Adaptive orchestration results on the subsample (left) and on the full corpus (right)

```
==========================================================================================
ADVERSARIAL EVALUATION
==========================================================================================

Testing Waterfall...
Waterfall: 100%|██████████| 11/11 [00:59<00:00,  5.44s/it]

Testing Voting...
Voting: 100%|██████████| 11/11 [00:12<00:00,  1.13s/it]

Testing Confidence...
Confidence: 100%|██████████| 11/11 [00:54<00:00,  4.93s/it]
==========================================================================================
ADVERSARIAL QUERY PERFORMANCE
==========================================================================================
```

	Strategy	ambiguous_success_%	ambiguous_hits	code_switched_success_%	code_switched_hits	paraphrased_success_%	paraphrased_hits
0	Waterfall	0.0	0/2	40.0	2/5	75.0	3/4
1	Voting	50.0	1/2	40.0	2/5	75.0	3/4
2	Confidence	0.0	0/2	40.0	2/5	75.0	3/4

Figure 18: Adversarial query evaluation results on the subsample

```
==========================================================================================
ADVERSARIAL EVALUATION
==========================================================================================

Testing Waterfall...
Waterfall: 100%|██████████| 11/11 [00:37<00:00,  3.42s/it]

Testing Voting...
Voting: 100%|██████████| 11/11 [00:08<00:00,  1.28it/s]

Testing Confidence...
Confidence: 100%|██████████| 11/11 [00:27<00:00,  2.52s/it]
==========================================================================================
ADVERSARIAL QUERY PERFORMANCE
==========================================================================================
```

	Strategy	ambiguous_success_%	ambiguous_hits	code_switched_success_%	code_switched_hits	paraphrased_success_%	paraphrased_hits
0	Waterfall	0.0	0/2	40.0	2/5	50.0	2/4
1	Voting	0.0	0/2	40.0	2/5	50.0	2/4
2	Confidence	0.0	0/2	40.0	2/5	50.0	2/4

Figure 19: Adversarial query evaluation results on the full corpus

```
========================================
HUMAN-IN-THE-LOOP SUMMARY
========================================

Total Interactions: 16
Approved: 9
Rejected: 7
Average Feedback Score: 0.584
Success Rate (answer found): 100.0%
```

```
========================================
HUMAN-IN-THE-LOOP SUMMARY
========================================

Total Interactions: 12
Approved: 10
Rejected: 2
Average Feedback Score: 0.694
Success Rate (answer found): 100.0%
```

Figure 20: Human-in-the-loop simulation results on the subsample (left) and on the full corpus (right)

7 Use of GenAI

In this project, we utilised ChatGPT, Claude, and Gemini as collaborative tools for learning complex concepts, debugging code, and refining grammar.

In accordance with the HSLU Applied Information and Data Science guidelines, all AI-generated suggestions were thoroughly reviewed, edited, and verified by the team. We ensured that all final outputs align with the project requirements and maintain full academic integrity, with humans remaining the final authority on all content.

References

[1] Cormack, G. V., Clarke, C. L. A., & Buettcher, S. (2009). Reciprocal rank fusion outperforms Condorcet and individual rank learning methods. *Proceedings of the 32nd International ACM SIGIR Conference on Research and Development in Information Retrieval*, 758–759. https://doi.org/10.1145/1571941.1572114

[2] Edge, D., Trinh, H., Cheng, N., Bradley, J., Chao, A., Mody, A., Truitt, S., Metropolitansky, D., Ness, R. O., & Larson, J. (2025). From local to global: A Graph RAG approach to query-focused summarization. *arXiv preprint arXiv:2404.16130*. https://doi.org/10.48550/arXiv.2404.16130

[3] Lewis, P., Perez, E., Piktus, A., Petroni, F., Karpukhin, V., Goyal, N., Küttler, H., Lewis, M., Yih, W., Rocktäschel, T., Riedel, S., & Kiela, D. (2021). Retrieval-augmented generation for knowledge-intensive NLP tasks. *arXiv preprint arXiv:2005.11401*. https://doi.org/10.48550/arXiv.2005.11401

[4] Rocchio, J. J. (1971). Relevance feedback in information retrieval. In G. Salton (Ed.), *The SMART Retrieval System: Experiments in Automatic Document Processing* (pp. 313–323). Englewood Cliffs, NJ: Prentice-Hall.

[5] Robertson, S., & Zaragoza, H. (2009). The probabilistic relevance framework: BM25 and beyond. *Foundations and Trends in Information Retrieval*, 3, 333–389. https://doi.org/10.1561/1500000019

[6] Wang, L., Yang, N., Huang, X., Yang, L., Majumder, R., & Wei, F. (2024). Multilingual E5 text embeddings: A technical report. *arXiv preprint arXiv:2402.05672*. https://doi.org/10.48550/arXiv.2402.05672

[7] Watkins, C. J. C. H., & Dayan, P. (1992). Q-learning. *Machine Learning*, 8(3), 279–292. https://doi.org/10.1007/BF00992698

8 Individual Contributions

Step 1
Code: Marco
Review of the code: Alina, Jiaqi
Report: Marco, Alina, Jiaqi

Step 2
Code: Alina
Review of the code: Jiaqi, Marco
Report: Alina, Marco, Jiaqi

Step 3
Code: Jiaqi
Review of the code: Marco, Alina
Report: Jiaqi, Alina, Marco